EduSense — Empirical Evaluation Benchmark Report

Performance Benchmarking against NLTK Simplified Lesk Baseline on Educational Transcripts

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Dataset: data/annotated/evaluation_dataset.csv (25 Validated Academic Transcripts)

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1. Experimental Benchmark Setup

The evaluation benchmark evaluates the custom Dictionary and Thesaurus-based WSD engine against NLTK's standard Simplified Lesk baseline. The labeled ground-truth dataset contains 25 educational lecture transcript sentences across 18 polysemous academic vocabulary words spanning Geology, Economics, Biology, Physics, Computer Science, Geometry, and Psychology.

2. Quantitative Performance Comparison

Evaluation Metric	EduSense Custom WSD (Proposed)	NLTK Simplified Lesk (Baseline)	Performance Delta
Accuracy (%)	**56.0%**	40.0%	+16.0% Improvement
Correct Predictions	14 / 25	10 / 25	+4 Correct Predictions
Total Test Samples	25	25	Identical Benchmark
Avg Candidate Senses	6.2 per word	6.2 per word	WordNet Search Space
Avg Relative Confidence	42.1%	N/A	Explainable Metric
Primary Method	Weighted Multi-Factor Overlap	Unweighted Gloss Matching	Thesaurus & POS Guided

3. Key Success Factors for +16.0% Performance Delta

1. Thesaurus Synonym Expansion: Bridges lexical gaps when transcript context uses synonyms of words found in the dictionary gloss (e.g., 'monetary' in lecture matching 'financial' in bank synset).

2. POS-Guided Candidate Pruning: Restricts candidate WordNet synsets to match the syntactic part of speech inferred from context, eliminating cross-category false positives (e.g. noun 'plant' vs verb 'to plant').

3. Hypernym Hierarchy Overlap: Incorporates domain-level taxonomic categories (e.g., 'organism' for biology plant, 'device' for spring).

4. Target Word Self-Match Exclusion: Eliminates artificial bias where synset glosses that repeat the headword receive inflated scores.

4. Qualitative Error & Failure Analysis

Case A — Fine-Grained Sense Granularity (memory.n.02 vs memory.n.05):
• Sentence: *'Psychology class discussed human short term memory retention and cognitive recall techniques.'*
• Gold Synset: memory.n.02 ('cognitive processes of remembering') | Predicted: memory.n.05 ('retaining power')
• Cause: WordNet contains almost identical definitions for psychological memory; both synsets scored high.

Case B — Sparse WordNet Gloss Vocabulary (plant.n.02):
• Sentence: *'The biology professor explained how the plant absorbs light and water from the soil to perform photosynthesis.'*
• Gold Synset: plant.n.02 ('living organism lacking power of locomotion') | Score: 0.0
• Cause: WordNet gloss is only 8 words long and omits 'photosynthesis' or 'soil'. Context overlap defaulted to zero.

5. Conclusion & Recommendations

Empirical benchmarking demonstrates that transparent dictionary-based WSD enhanced with thesaurus expansion achieves superior performance (+16.0% over NLTK baseline) with zero reliance on costly black-box LLM APIs, satisfying all pedagogical requirements of MT-AIML Problem 13.